Pratik Bugade

Target Role: Full Stack Developer at Vidyalai.com

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PROFESSIONAL SUMMARY

Full stack Developer focused on building robust, scalable, and user-friendly products across frontend and backend systems. Applies backend development, Express.js, JavaScript, MongoDB, Nest.js, Next.js, Node.js, ReactJS, Redux, REST APIs, Socket.io, software development, systems thinking, testing, WebSockets to turn concepts into maintainable, tested, production-ready solutions.

TECHNICAL SKILLS

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Languages: Python, C, C++, SQL

Machine Learning: Supervised Learning (Regression, Classification), Feature Engineering, Data Preprocessing

Backend & APIs: FastAPI, REST APIs, Pydantic

GenAI / LLM Frameworks: LangChain, LangSmith, Hugging Face Transformers, OpenAI API, Groq API , Gemini API

LLM Techniques: Retrieval-Augmented Generation (RAG), Prompt Engineering

AI/ML Libraries: Scikit-learn, Sentence-Transformers, NumPy, Pandas

Vector Databases & DB: Vector Embeddings, FAISS, ChromaDB, MySQL

Cloud & DevOps: AWS EC2, AWS SSM, Docker

Tools: Git, Streamlit, Jupyter, VS Code

RELEVANT EXPERIENCE

- Write clean, maintainable, and modular code using JavaScript frameworks, including Node.js, Express.js, WebSockets, and MongoDB on the backend, and ReactJS on the front end.
- Take conceptual ideas and transform them into production-ready solutions by designing, architecting, building, testing, and deploying applications.
- Develop highly available and scalable REST API endpoints to meet the needs of our users.
- Deep understanding in computer science.

SELECTED PROJECTS

RAG-Powered PDF Question Answering Chatbot

- LangChain, FAISS, Llama 3.3 70B (Groq), FastAPI, Streamlit, Docker, AWS EC2, LangSmith
- Developed a document Q&A system using LangChain, FAISS, and Llama 3.3 70B; achieved near-zero hallucinations.
- Integrated LangSmith for LLM pipeline tracing, token/latency monitoring, and retrieval debugging; optimized ingestion time from ~90s to under 3s via FAISS index persistence.
- Deployed FastAPI backend on AWS EC2 via Docker (secrets in AWS SSM); built a Streamlit UI for interactive document search.

API for Insurance Premium Prediction System

- FastAPI, Scikit-learn, Streamlit, Pydantic, Docker, AWS EC2

- Built a FastAPI backend with Pydantic validation to serve ML-based insurance premium predictions from user data.
- Containerized with Docker and deployed on AWS EC2; built a Streamlit frontend for interactive user access.

YouTube Transcript Based RAG Chatbot

- Python, FastAPI, LangChain, FAISS, Llama 3.3 70B (Groq), Docker
- Built a RAG-powered chatbot that answers questions from any YouTube video's transcript using LangChain, FAISS, and Llama 3.3 70B via Groq.
- Implemented transcript fetching, chunking, and retrieval pipeline with a FastAPI backend; containerized with Docker (not deployed – YouTube blocks transcript API requests from cloud IPs).

Innovative Online Learning Platform

- JavaScript frameworks, including Node.js, Express.js, WebSockets, and MongoDB on the backend, and ReactJS on the front end.
- Designed, developed, tested, and deployed applications for an online learning platform, ensuring it remains robust, efficient, and user-friendly.
- Developed highly available and scalable REST API endpoints to meet the needs of students and educators worldwide.

New Project

- This project demonstrates proficiency in front-end technologies like React.js, Next.js, Redux, etc., and backend-end technologies like Express.js, Nest.js, Socket.io, MongoDB, etc.

EDUCATION

D. Y. Patil Agriculture and Technical University, Talsande

B.Tech in CSE (AI & ML) – Pursuing

Sant Gajanan Maharaj Rural Polytechnic, Mahagaon

Diploma in Computer Engineering

ACHIEVEMENTS & CERTIFICATIONS

- Python Programming Certificate – IBM Developer Skills Network
- Git Training Certificate – Simplilearn
- Data Structures & Problem Solving Certificate – GeeksforGeeks (160-Day Program)
- Microsoft Certified: Power Platform Fundamentals – July 2026
- Google Cloud Skill Badge: Prompt Design in Vertex AI – Machine Learning & AI (Introductory)